

QUESTION PAPER

Name of the Examination: WINTER 2022-2023 – CAT-2

Course Code: CSE1008

Course Title: Theory of Computation

Set number: 01

Date of Exam: 28/03/2023 (AN) (B₂)

Duration: 90 minutes

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. Answer the following.

(5 + 5 = 10 Marks)

- (a). Show that the language $L = \{0^k 10^k \mid k \geq 0\}$ is not a regular language. **(5 Marks)**
(b). Construct a CFG(Context Free Grammar) for the language $L = \{a^n b^{2n} \mid n \geq 1\}$. **(5 Marks)**
Write the tuple form for CFG.

Q2. Consider G whose productions P are:

(10 Marks)

- $S \rightarrow A1B$
 $A \rightarrow 0A \mid \epsilon$
 $B \rightarrow 0B \mid 1B \mid \epsilon$

For the string $w = '00101'$, find the Leftmost and Rightmost derivation and also construct both Leftmost parse tree and Rightmost Parse tree for the yield = '00101'.

Q3. Check that the given grammar is ambiguous or not. Write derivations for the string $a+a*a$ and then check for ambiguity.

(10 Marks)

- $S \rightarrow S + S$
 $S \rightarrow S * S$
 $S \rightarrow S \wedge S$
 $S \rightarrow a$

Q4. Construct a PDA (Push Down Automata) for the Language, $L = \{a^n b^m c^n \mid n \geq 1, m \geq 1\}$.

Give ID(Instantaneous Description) representations for the string 'abbc'. (10 Marks)

Q5. Convert the following CFG(Context Free Grammar) to a PDA(Push Down Automata) that accepts the language by empty stack: Write the tuple form.

$$S \rightarrow OS1 \mid A$$

$$A \rightarrow 1A0 \mid S \mid \epsilon$$

Test whether the input string '0101' is acceptable by the PDA or not? **(10 Marks)**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	2,3	2	1,2,3	1,4	1,2,3	10 (5 + 5)
Q2	3	2	1,2,3	1,4	1,2,3	10
Q3	3	3	3,4,5	1,2,3,4	1,2,3	10
Q4	4	3	3,4,5	1,2,3,4	1,2,3	10
Q5	4	3	3,4,5	1,2,3,4	1,2,3	10

QUESTION PAPER

Name of the Examination: WINTER 2022-2023 – CAT-2

Course Code: CSE1008

Course Title: Theory of Computation

Set number: 02

Date of Exam: 27/03/2023 (AN)

Duration: 90 minutes

Total Marks: 50

(AN)

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. Answer the following.

(5 + 5 = 10 Marks)

(a). Prove that the language $L = \{0^n 1^m \mid n \leq m\}$ is not a regular language. (5 Marks)

(b). Design context free grammar for the language $L = \{w \in \{0, 1\}^* \mid \text{the length of } w \text{ is odd and the middle symbol is } 1\}$. Write the tuple form for CFG. (5 Marks)

Q2. Consider G whose productions P are:

(10 Marks)

$S \rightarrow aB \mid bA$
 $S \rightarrow a \mid aS \mid bAA$
 $S \rightarrow b \mid aS \mid aBB$

For the string $w = 'aabbabba'$, find the Leftmost and Rightmost derivation and also construct both Leftmost parse tree and Rightmost Parse tree for the yield = 'aabbabba'.

Q3. Check that the given grammar is ambiguous or not. Write derivations for the string a(a)aa and then check for ambiguity. (10 Marks)

$A \rightarrow AA$

$A \rightarrow (A)$

$A \rightarrow a$

Q4. Construct a PDA for language $L = \{ww^R \mid w \in \{0, 1\}^*\}$ where w^R is the reverse of w .

Specify all transitions. Give ID (Instantaneous Description) representations for the string

'abbbba'.

(10 Marks)

Q5. Convert the following CFG (Context Free Grammar) to a PDA (Push Down Automata) that accepts the language by empty stack: Write the tuple form. (10 Marks)

$S \rightarrow aSb$

$S \rightarrow a \mid b \mid \epsilon$

Test whether the input string 'aaabb' is acceptable by the PDA or not?

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	2,3	2	1,2,3	1,4	1,2,3	10 (5 + 5)
Q2	3	2	1,2,3	1,4	1,2,3	10
Q3	3	3	3,4,5	1,2,3,4	1,2,3	10
Q4	4	3	3,4,5	1,2,3,4	1,2,3	10
Q5	4	3	3,4,5	1,2,3,4	1,2,3	10

QUESTION PAPER

Name of the Examination: WINTER 2022-2023 – CAT-2

Course Code: CSE1008

Course Title: Theory of Computation

Set number: 05

Date of Exam:

30/03/2023 (AN)

Duration: 90 minutes

Total Marks: 50

(02)

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. Answer the following.

(5 + 5 = 10 Marks)

(a) Prove that the language $L = \{a^p \mid p \text{ is prime}\}$ is not a regular language.

(5 Marks)

(b) Design context free grammar for the language $\{w \in \{0, 1\}^* \mid w = w^R \text{ and } |w| \text{ is even}\}$. Write the tuple form for CFG.

(5 Marks)

Q2. Consider G whose productions P are:

(10 Marks)

$S \rightarrow AB \mid \epsilon$

$A \rightarrow aB$

$B \rightarrow Sb$

For the string $w = 'aabbbb'$, find the Leftmost and Rightmost derivation and also construct both Leftmost parse tree and Rightmost Parse tree for the yield = 'aabbbb'.

Q3. Check that the given grammar is ambiguous or not. Write derivations for the string 'abababb' and then check for ambiguity.

(10 Marks)

$S \rightarrow a / abSb / aAb$

$A \rightarrow bS / aAAb$

Q4. Construct a PDA for language $L = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i + k = j\}$. Specify all transitions.

Give ID (Instantaneous Description) representations for the string 'abbbcc'.

(10 Marks)

Q5. Convert the following CFG (Context Free Grammar) to a PDA (Push Down Automata) that accepts the language by empty stack: Write the tuple form. (10 Marks)

$S \rightarrow aSbbb$

$S \rightarrow \epsilon$

Test whether the input string 'aabbbbbb' is acceptable by the PDA or not?

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	2,3	2	1,2,3	1,4	1,2,3	10 (5 + 5)
Q2	3	2	1,2,3	1,4	1,2,3	10
Q3	3	3	3,4,5	1,2,3,4	1,2,3	10
Q4	4	3	3,4,5	1,2,3,4	1,2,3	10
Q5	4	3	3,4,5	1,2,3,4	1,2,3	10